

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250283419

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Hafner; Matthew Troy et al.

COOLING CIRCUIT FOR A STATOR VANE BRAZE JOINT

Abstract

A stator vane includes a platform that defines an opening. The stator vane further includes an airfoil that has a leading edge, a trailing edge, a suction side wall, and a pressure side wall. The airfoil extends radially between a base and a tip. At least one of the base or the tip includes a protrusion. The protrusion extends into the opening of the platform such that the platform surrounds the protrusion of the airfoil. The stator vane further includes a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil. The stator vane further includes a cooling circuit defined in at least one of the protrusion or the platform to cool the braze joint.

Inventors: Hafner; Matthew Troy (Honea Path, SC), Matthews; John M. (Greer, SC), Snider; Zachary John (Simpsonville, SC)

Applicant: GE Vernova Infrastructure Technology LLC (Greenville, SC)

Family ID: 1000008629215

Appl. No.: 19/220165

Filed: May 28, 2025

Related U.S. Application Data

parent US division 18591666 20240229 parent-grant-document US 12338749 child US 19220165

Publication Classification

Int. Cl.: F01D9/04 (20060101); F01D5/18 (20060101)

U.S. Cl.:

CPC F01D9/044 (20130101); F01D5/187 (20130101); F01D5/188 (20130101); F01D5/189 (20130101); F01D9/041 (20130101); F05D2230/237 (20130101); F05D2240/81

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a divisional application of U.S. application Ser. No. 18/591,666 filed Feb. 29, 2024, which is a non-provisional application claiming the benefit of priority under 35 U.S.C. § 120 to U.S. Non-Provisional application Ser. No. 17/868,911, filed Jul. 20, 2022, which is hereby incorporated by reference in its entirety.

FIELD

[0002] The present disclosure relates generally to a stator vane for a turbomachine having one or more components joined with a braze joint. Particularly, the present disclosure relates to a cooling circuit for cooling the braze joint of the stator vane.

BACKGROUND

[0003] Turbomachines are utilized in a variety of industries and applications for energy transfer purposes. For example, a gas turbine engine generally includes a compressor section, a combustion section, a turbine section, and an exhaust section. The compressor section progressively increases the pressure of a working fluid entering the gas turbine engine and supplies this compressed working fluid to the combustion section. The compressed working fluid and a fuel (e.g., natural gas) mix within the combustion section and burn in a combustion chamber to generate high pressure and high temperature combustion gases. The combustion gases flow from the combustion section into the turbine section where they expand to produce work. For example, expansion of the combustion gases in the turbine section may rotate a rotor shaft connected, e.g., to a generator to produce electricity. The spent combustion gases then exit the gas turbine via the exhaust section.

[0004] During operation of the turbomachine, various hot gas path components in the system are subjected to high temperature flows, which can cause the hot gas path components to fail. Since higher temperature flows generally result in increased performance, efficiency, and power output of the turbomachine, the hot gas path components that are subjected to high temperature flows must be cooled to allow the gas turbine system to operate with flows at increased temperatures.

[0005] As the maximum local temperature of the hot gas path components approaches the melting temperature of the hot gas path components, forced air cooling becomes necessary. For this reason, airfoils of turbine rotor blades and stationary nozzles often require complex cooling schemes in which air, typically bleed air from the compressor section, is forced through internal cooling passages within the airfoil and then discharged through cooling holes at the airfoil surface to transfer heat from the hot gas path component.

[0006] Many complex cooling schemes use small cooling passages, or micro-channels, to deliver cooling fluid through the airfoil. Such cooling schemes present a considerable fabrication challenge for cores and castings, which can significantly increase the manufacturing cost of the hot gas path components using such known near wall cooling systems. To address the fabrication challenges with complex and/or small cooling channels near the component surface, many hot gas path components with such features may be additively manufactured. Additive manufacturing is capable of producing components with intricate and varied cooling features. However, additively manufacturing a hot gas path component, such as a rotor blade or stator vane, as a single component may be costly and time-consuming. Additionally, manufacturing errors in a single portion of the hot gas path component may result in the scrapping of the entire component.

[0007] As such, manufacturing a hot gas path component as multiple sub-components may be advantageous. However, due to the complex geometries of the sub-components, joining the sub-components to form the hot gas path component may be difficult. Additionally, the joints formed

between the sub-components may be particularly weak and/or fail when exposed to the hot combustion gases produced during operation of the turbomachine and transmitted through the hot gas path through the turbine section.

[0008] Accordingly, an improved hot gas path component, having one or more subcomponents joined together and capable of being subject to hot combustion gases without risk of joint failure, is desired and would be appreciated in the art.

BRIEF DESCRIPTION

[0009] Aspects and advantages of the stator vanes and turbomachines in accordance with the present disclosure will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the technology.

[0010] In accordance with one embodiment, a stator vane is provided. The stator vane includes a platform that defines an opening. The stator vane further includes an airfoil that has a leading edge, a trailing edge, a suction side wall, and a pressure side wall. The airfoil extends radially between a base and a tip. At least one of the base or the tip includes a protrusion. The protrusion extends into the opening of the platform such that the platform surrounds the protrusion of the airfoil. The stator vane further includes a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil. The stator vane further includes a cooling circuit defined in one of the protrusion or the platform to cool the braze joint.

[0011] In accordance with another embodiment, a turbomachine is provided. The turbomachine includes a compressor section, a combustion section, and a turbine section. At least one stator vane is disposed in the turbine section. The at least one stator vane includes a platform that defines an opening. The stator vane further includes an airfoil that has a leading edge, a trailing edge, a suction side wall, and a pressure side wall. The airfoil extends radially between a base and a tip. At least one of the base or the tip includes a protrusion. The protrusion extends into the opening of the platform such that the platform surrounds the protrusion of the airfoil. The stator vane further includes a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil. The stator vane further includes a cooling circuit defined in one of the protrusion or the platform to cool the braze joint.

[0012] These and other features, aspects, and advantages of the present stator vanes and turbomachines will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the technology and, together with the description, serve to explain the principles of the technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] A full and enabling disclosure of the present stator vanes and turbomachines, including the best mode of making and using the present systems and methods, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:

[0014] FIG. 1 is a schematic illustration of a turbomachine (e.g., a gas turbine engine), in accordance with embodiments of the present disclosure;

[0015] FIG. 2 illustrates a partial cross-sectional side view of a turbine section of a gas turbine engine as shown in FIG. 1, in accordance with embodiments of the present disclosure;

[0016] FIG. 3 illustrates an exploded view of a stator vane, in accordance with embodiments of the present disclosure;

[0017] FIG. 4 illustrates an overhead perspective view of a stator vane, in accordance with embodiments of the present disclosure;

[0018] FIG. 5 illustrates a cross-sectional perspective view of a stator vane, in accordance with

embodiments of the present disclosure;

[0019] FIG. 6 schematically illustrates a cross-sectional planar view of the stator vane of FIG. 5, in accordance with embodiments of the present disclosure;

[0020] FIG. 7 schematically illustrates a cross-sectional planar view of a stator vane, in accordance with embodiments of the present disclosure;

[0021] FIG. 8 schematically illustrates a cross-sectional planar view of a stator vane from along the line 8-8 shown in FIG. 7, in accordance with embodiments of the present disclosure;

[0022] FIG. 9 schematically illustrates a cross-sectional planar view of a stator vane, in accordance with embodiments of the present disclosure;

[0023] FIG. 10 illustrates a perspective view of a portion of a stator vane, in accordance with embodiments of the present disclosure;

[0024] FIG. 11 illustrates a cross-sectional perspective view of a stator vane, in accordance with embodiments of the present disclosure;

[0025] FIG. 12 illustrates an enlarged cross-sectional planar view of the stator vane of FIG. 11, in accordance with embodiments of the present disclosure;

[0026] FIG. 13 illustrates a perspective view of a portion of a stator vane, in accordance with embodiments of the present disclosure;

[0027] FIG. 14 illustrates an enlarged cross-sectional planar view of the stator vane of FIG. 13, in accordance with embodiments of the present disclosure;

[0028] FIG. 15 illustrates an overhead perspective view of a stator vane in accordance with embodiments of the present disclosure;

[0029] FIG. 16 schematically illustrates an enlarged cross-sectional planar side view of a stator vane, in accordance with embodiments of the present disclosure; and

[0030] FIG. 17 schematically illustrates an enlarged cross-sectional planar side view of a stator vane, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

[0031] Reference now will be made in detail to embodiments of the present stator vanes and turbomachines, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation, rather than limitation of, the technology. In fact, it will be apparent to those skilled in the art that modifications and variations can be made in the present technology without departing from the scope or spirit of the claimed technology. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present disclosure covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0032] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other implementations. Additionally, unless specifically identified otherwise, all embodiments described herein should be considered exemplary.

[0033] The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the invention. As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

[0034] The term “fluid” may be a gas or a liquid. The term “fluid communication” means that a fluid is capable of making the connection between the areas specified.

[0035] As used herein, the terms “upstream” (or “forward”) and “downstream” (or “aft”) refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows. The term “radially” refers to the relative direction that is substantially perpendicular to

an axial centerline of a particular component; the term “axially” refers to the relative direction that is substantially parallel and/or coaxially aligned to an axial centerline of a particular component; and the term “circumferentially” refers to the relative direction that extends around the axial centerline of a particular component.

[0036] Terms of approximation, such as “about,” “approximately,” “generally,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 1, 2, 4, 5, 10, 15, or 20 percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values. When used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction. For example, “generally vertical” includes directions within ten degrees of vertical in any direction, e.g., clockwise or counter-clockwise.

[0037] The terms “coupled,” “fixed,” “attached to,” and the like refer to both direct coupling, fixing, or attaching, as well as indirect coupling, fixing, or attaching through one or more intermediate components or features, unless otherwise specified herein. As used herein, the terms “comprises,” “comprising,” “includes,” “including,” “has,” “having” or any other variation thereof, are intended to cover a non-exclusive inclusion. For example, a process, method, article, or apparatus that comprises a list of features is not necessarily limited only to those features but may include other features that are not expressly listed or that are inherent to such process, method, article, or apparatus. Further, unless expressly stated to the contrary, “or” refers to an inclusive-or and not to an exclusive-or. For example, a condition “A or B” is satisfied by any one of the following: A is true (or present) and B is false (or not present). A is false (or not present) and B is true (or present), and both A and B are true (or present).

[0038] Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

[0039] Referring now to the drawings, FIG. **1** illustrates a schematic diagram of one embodiment of a turbomachine, which in the illustrated embodiment is a gas turbine engine **10**. Although an industrial or land-based gas turbine is shown and described herein, the present disclosure is not limited to an industrial or land-based gas turbine engine, unless otherwise specified in the claims. For example, the invention as described herein may be used in any type of turbomachine including, but not limited to, a steam turbine, an aircraft gas turbine, or a marine gas turbine.

[0040] As shown in FIG. **1**, the gas turbine **10** generally includes a compressor section **12**. The compressor section **12** includes a compressor **14**. The compressor includes an inlet **16** that is disposed at an upstream end of the gas turbine engine **10**. The gas turbine engine **10** further includes a combustion section **18** having one or more combustors **20** disposed downstream from the compressor section **12**. The gas turbine engine **10** further includes a turbine section **22** (i.e., an expansion turbine) that is downstream from the combustion section **18**. A shaft **24** extends generally axially through the gas turbine engine **10**.

[0041] The compressor section **12** may generally include a plurality of rotor disks **21** and a plurality of rotor blades **23** extending radially outwardly from and connected to each rotor disk **21**. Each rotor disk **21** in turn may be coupled to or form a portion of the shaft **24** that extends through the compressor section **12**. The rotor blades **23** of the compressor section **12** may include turbomachine airfoils that define an airfoil shape (e.g., having a leading edge, a trailing edge, and side walls extending between the leading edge and the trailing edge).

[0042] The turbine section **22** may generally include a plurality of rotor disks **27** and a plurality of rotor blades **28** extending radially outwardly from and being interconnected to each rotor disk **27**. Each rotor disk **27** in turn may be coupled to or form a portion of the shaft **24** that extends through the turbine section **22**. The turbine section **22** further includes an outer casing **32** that circumferentially surrounds the portion of the shaft **24** and the rotor blades **28**. The turbine section **22** may include stator vanes or stationary nozzles **26** extending radially inward from the outer casing **32**. The rotor blades **28** and stator vanes **26** may be arranged in alternating stages along an axial centerline **30** of gas turbine **10**. Both the rotor blades **28** and the stator vanes **26** may include turbomachine airfoils that define an airfoil shape (e.g., having a leading edge, a trailing edge, and side walls extending between the leading edge and the trailing edge).

[0043] In operation, ambient air **36** or other working fluid is drawn into the inlet **16** of the compressor **14** and is progressively compressed to provide a compressed air **38** to the combustion section **18**. The compressed air **38** flows into the combustion section **18** and is mixed with fuel to form a combustible mixture. The combustible mixture is burned within a combustion chamber **40** of the combustor **20**, thereby generating combustion gases **42** that flow from the combustion chamber **40** into the turbine section **22**. Energy (kinetic and/or thermal) is transferred from the combustion gases **42** to the rotor blades **28**, causing the shaft **24** to rotate and produce mechanical work. The spent combustion gases **42** (also called “exhaust gases”) exit the turbine section **22** and flow through the exhaust diffuser **34** across a plurality of struts or main airfoils **44** that are disposed within the exhaust diffuser **34**.

[0044] The gas turbine **10** may define a cylindrical coordinate system having an axial direction A extending along the axial centerline **30**, a radial direction R perpendicular to the axial centerline **30**, and a circumferential direction C extending around the axial centerline **30**.

[0045] FIG. **2** is a partial cross-sectional side view of the turbine section **22** of the gas turbine engine **10**, in accordance with embodiments of the present disclosure. The turbine section **22** may include one or more stages **50** that each include a set of rotor blades **28** coupled to a rotor disk **27** that may be rotatably attached to the shaft **24**. The one or more stages **50** may further include a set of stator vanes **26**. The stator vane **26** described herein may be employed in a first stage, a second stage, a third stage, or combinations thereof, in which “first” refers to the stage immediately downstream of the combustion section **18**.

[0046] Each stator vane **26** may include at least one airfoil **56** that extends in the radial direction R between an inner platform or endwall **52** and an outer platform or endwall **54**. The circumferentially adjacent outer platforms **54** of each stator vane **26** may be coupled together to form an outer annular ring extending around an inner annular ring of the circumferentially adjacent inner platforms **52** of each stator vane **26**. The at least one airfoil **56** may extend between the two annular rings formed by the platforms **52**, **54**. The turbine section **22** may also include shroud segments **58**, which may be disposed downstream of the outer platform **54** to direct combustion gases **42** flowing past the stator vanes **26** to the rotor blades **28**.

[0047] Structures or components disposed along the flow path of the combustion gases **42** may be referred to as hot gas path components. In one example, the hot gas path component may be the stator vane **26** and/or the rotor blade **28**. In some embodiments, to cool the hot gas path components, cooling features, such as impingement sleeves, cooling channels, cooling holes, etc. may be disposed within the hot gas path components, as indicated by the dashed line **78**. For example, cooling air as indicated by an arrow **79** may be routed from the compressor section **12** or elsewhere and directed through the cooling features as indicated by arrows **81**.

[0048] FIG. **3** illustrates an exploded view of a stator vane **26**, in which the inner platform **52**, the outer platform **54**, and the airfoil are separated from one another along an axial centerline **55** of the stator vane **26**, in accordance with embodiments of the present disclosure. The axial centerline **55** may be generally parallel to the radial direction of the gas turbine **10** when the stator vane **26** is installed in the turbine section **22**. As shown, the stator vane **26** may include an inner platform **52**,

an outer platform **54**, and an airfoil **56** extending between the inner platform **52** and the outer platform **54**. The airfoil **56** may define a generally aerodynamic shape or contour. For example, the airfoil **56** may include a leading edge **60** that engages a flow of combustion gases and side walls that guide the combustion gases along the airfoil **56** to a trailing edge **62**. For example, the airfoil **56** may include a pressure side wall **64** and a suction side wall **66** that each extend from the leading edge **60** to the trailing edge **62**. Additionally, the airfoil **56** may extend radially between a base **68** and a tip **70**. The base **68** is configured to be coupled to the inner platform **52**, and the tip **70** is configured to be coupled to the outer platform **54**.

[0049] In exemplary embodiments, the inner platform **52**, the airfoil **56**, and the outer platform **54** may be separate components (e.g., manufactured as separate components) that are brazed or welded to one another via one or more braze joints. For example, in particular embodiments, the inner platform **52**, the airfoil **56**, and the outer platform **54** may each be separately additively manufactured (e.g., 3D printed) and subsequently joined to one another via braze joints. Forming the stator vane **26** as three separate components advantageously increases the repairability of the stator vane **26**. For example, if a portion of the airfoil **56** is damaged, then the entire stator vane **26** would not need to be replaced. Rather, the braze connection could be undone (e.g., via reheating and melting the braze joint) to decouple the airfoil **56** from the platforms **52**, **54**, and a new airfoil **56** could be employed or the old airfoil could be repaired.

[0050] Additionally, in instances where additive manufacturing is used to produce some portion of or all of the stator vane **26**, the build (i.e., print) time of the airfoil **56** as a separate component from the inner platform **52** and the outer platform **54** is significantly shorter as compared to the build time of an integral nozzle in which the inner and outer platforms **52**, **54** are printed with the airfoil **56**. Moreover, forming the stator vane **26** from three separate components permits different manufacturing techniques and/or different materials to be used for the various components.

[0051] As shown in FIG. 3, the airfoil **56** may include a camber axis **57**, which may extend from the leading edge **60** to the trailing edge **62** and which may be defined halfway between the pressure side wall **64** and the suction side wall **66**. The camber axis **57** may be curved and/or contoured to correspond with the curve of the pressure side wall **64** and the suction side wall **66**. A transverse direction T may be defined orthogonally with respect to the camber axis **57**. As shown in FIG. 3, the stator vane **26** may further include one or more protrusions **72** that extend outwardly from the airfoil **56** in both the radial direction R and in the transverse direction T. For example, the stator vane **26** may include a first protrusion **72** that extends from the airfoil **56** at the base **68** and a second protrusion **72** that extends from the airfoil **56** at the tip **70**. Each protrusion **72** may extend generally perpendicularly from the pressure side wall **64** and the suction side wall **66** in the transverse direction T.

[0052] Both the inner platform **52** and the outer platform **54** may define an opening **102**. The opening **102** may be sized and shaped to correspond with the protrusion **72**, such that the protrusion **72** may be inserted into each opening **102** (and subsequently brazed to the respective platform), thereby coupling the airfoil **56** to the inner platform **52** and the outer platform **54**. Particularly, the protrusion **72** extending from the base **68** of the airfoil **56** may be inserted into the opening **102** of the inner platform **52** (and subsequently brazed to the inner platform **52**), and the protrusion **72** of extending from the tip **70** of the airfoil **56** may be inserted into the opening **102** of the outer platform **54** (and subsequently brazed to the outer platform **54**).

[0053] For example, the protrusions **72** of the airfoil **56** may be brazedly coupled to the inner and outer platforms **52** and **54** via a braze joint **106** (FIGS. 4 and 5). That is, as shown in FIG. 6, a braze material **107** may be disposed between the protrusion **72** and the boundary of the opening **102** (at both the inner and outer platforms **52** and **54**). The entire stator vane **26** may then be placed in a braze oven to melt the braze material **107**, and subsequently, the braze material **107** may solidify thereby joining the inner and outer platforms **52** and **54** to a respective protrusion **72** of the airfoil **56**. The braze material may contain copper, nickel, silver, gold, aluminum, or other suitable

brazed metals.

[0054] In many embodiments, as shown in FIG. 3, the inner platform 52 and/or the outer platform 54 includes a main body 53. The main body 53 may extend generally perpendicularly to the airfoil 56. The main body 53 of the inner platform 52 may define a radially inward flow boundary for combustion gases in the turbine section 22. Similarly, the main body 53 of the outer platform 54 may define a radially outer flow boundary for combustion gases in the turbine section 22.

Additionally, the main body 53 of the inner platform 52 and the outer platform 54 may at least partially define the respective openings 102. In exemplary embodiments, one or both of the inner platform 52 and/or the outer platform 54 may include a raised wall 80 extending radially from the main body 53. The raised wall 80 may at least partially define the opening 102. In many embodiments, the protrusion 72 may be brazedly coupled to the raised wall 80. In some embodiments, the inner and outer platforms 52 and 54 may include one or more rails 82 (which may extend across the main body 53 generally in the circumferential direction C).

[0055] In certain embodiments, the protrusion 72 that extends from the base 68 of the airfoil 56 may extend into the opening 102 of the inner platform 52 such that the inner platform 52 surrounds the protrusion 72 of the airfoil 56. Particularly, the main body 53 and/or the raised wall 80 of the inner platform 52 may surround the protrusion 72 extending from the base 68 of the airfoil 56 (i.e., the raised wall 80 extends around a perimeter of the protrusion 72). Similarly, the protrusion 72 that extends from the tip 70 of the airfoil 56 may extend into the opening 102 of the outer platform 54 such that the outer platform 54 surrounds the protrusion 72 of the airfoil 56. Particularly, the main body 53 and/or the raised wall 80 of the outer platform 54 may surround the protrusion 72 extending from the tip 70 of the airfoil 56.

[0056] In many embodiments, the protrusion 72 of the airfoil 56 may include a leading edge portion 84, a trailing edge portion 86, a pressure side portion 88, and a suction side portion 90. The leading edge portion 84, the pressure side portion 88, and the suction side portion 90 may at least partially define a cavity 92 that extends into and is further defined in the airfoil 56. The cavity 92 may be exposed by the opening 102 of the platform 54, such that air (e.g., bleed air from the compressor 14) may enter the cavity 92. A rib 94 (FIGS. 4, 10) may extend across the cavity 92, e.g., between the pressure side portion 88 and the suction side portion 90 of the protrusion 72, and between the pressure side wall 64 and the suction side wall 66 of the airfoil 56, to partition the cavity 92 into multiple portions and to provide structural support for the airfoil 56. The protrusion 72 may further include a solid tail 96, which may provide a flat, and/or smooth, surface for additional braze material to bond the protrusion 72 to the platform 54 (as shown by the grid in FIG. 4).

[0057] The leading edge portion 84 of the protrusion 72 may extend from the leading edge 60 of the airfoil 56. The trailing edge portion 86 of the protrusion 72 may extend from the trailing edge 62 of the airfoil 56. The pressure side portion 88 of the protrusion 72 may extend from the pressure side wall 64 of the airfoil 56. The suction side portion 90 of the protrusion 72 may extend from the suction side wall 66 of the airfoil 56.

[0058] FIG. 4 illustrates an enlarged perspective view of a stator vane 26, in accordance with embodiments of the present disclosure. As shown, the protrusion 72 of the airfoil 56 may extend into the opening 102 of the platform 54 such that the platform 54 surrounds the protrusion 72 of the airfoil 56. A braze joint 106 (as shown by the dashed lines arranged in grids) is disposed between and fixedly couples the platform 54 and the protrusion 72 of the airfoil 56.

[0059] FIGS. 5-9 each illustrate a cross-sectional view of a stator vane 26 in accordance with one or more embodiments of the present disclosure. Particularly, FIG. 5 may be a cross-sectional view of a stator vane 26 from along the line 5-5 shown in FIG. 4, and FIG. 6 may be an enlarged planar view of the stator vane 26 in FIG. 5. As shown, the stator vane 26 may include a platform 100 (such as the inner platform 52 or the outer platform 54 shown in FIG. 3). The platform 100 may define an opening 102, and an airfoil 56 may extend into the opening 102 and couple to the

platform **100**. A braze joint **106** may be disposed between the airfoil **56** and the platform **100**, thereby coupling the airfoil **56** to the platform **100**. The braze joint **106** may include a braze material **107** that bonds the platform **100** to the airfoil **56**. The braze material **107** may have a melting temperature that is lower than a melting temperature of a material from which the airfoil **56** and/or the platform **100** are formed. For example, the braze material **107** may be copper, nickel, silver, gold, aluminum, or combinations thereof. The airfoil **56** and the platform **100** may be formed from a metal material having a high melting temperature, such as nickel and cobalt alloys or other materials.

[0060] In exemplary embodiments, the stator vane **26** may include a cooling circuit **108** defined in one of the airfoil **56** and/or the platform **100** to cool the braze joint **106** (e.g., during operation of the gas turbine engine **10**). In some embodiments, the cooling circuit **108** may be defined in one or both of the protrusion **72** and/or the airfoil **56**. In other embodiments, the cooling circuit **108** may be defined in the platform **100**, such as in one or more of the main body **53**, the raised wall **80**, and/or the rail **82** described above with reference to FIG. **3**. The cooling circuit **108** may be in fluid communication with the compressor section **12**, such that the cooling circuit **108** may utilize a flow of compressed air from the compressor to cool the braze joint **106**.

[0061] In many embodiments, the platform **100** may include an inner wall **110** and an outer wall **112**, and the raised wall **80** may extend between the inner wall **110** and the outer wall **112**. In exemplary embodiments, as shown, the braze joint **106** may be disposed between, and fixedly couple, the raised wall **80** of the platform **100** to the protrusion **72** of the airfoil **56**. In many embodiments, an inlet plenum **114** may be defined between the inner wall **110**, the outer wall **112**, and the raised wall **80**. One or more inlet apertures **116** may be defined in the outer wall **112** to allow compressed air to enter the inlet plenum **114** (e.g., from a compressed air source such as bleed air from the compressor section **12**). The inner wall **110** may define an outer surface **134** partially defining the inlet plenum **114** and an inner surface **136** partially defining a hot gas path boundary of the combustion gases. In other words, the inner surface **136** may be exposed to the combustion gases during operation of the gas turbine **10**.

[0062] In many embodiments, as shown in FIGS. **6** and **7**, the cooling circuit **108** may include one or more inlet channels **118**, a plenum **120**, and one or more outlet channels **122** each defined in the raised wall **80** and/or the inner wall **110** of the platform **100**. The one or more inlet channels **118** may extend from the inlet plenum **114** to a plenum **120**. In some embodiments, the plenum **120** may be annular, such that it surrounds the protrusion **72** (thereby surrounding the axial centerline **55** shown in FIG. **3**). Each inlet channel **118** of the one or more inlet channels **118** may be sized and oriented to direct coolant or air **150** to impinge upon a joint boundary wall **124**. The joint boundary wall **124** may form a portion of the raised wall **80** and may be disposed between the plenum **120** and the braze joint **106**. For example, the joint boundary wall **124** of the raised wall **80** may be oriented generally radially, and the inlet channel **118** may include an axially oriented portion immediately upstream of an outlet of the inlet channel **118**, such that coolant exits the inlet channel **118** traveling generally axially (or perpendicularly to the joint boundary wall **124**). The air **150** may then impinge upon, or strike, the joint boundary wall **124**, thereby transferring heat from the joint boundary wall **124** to the air **150**. Because the joint boundary wall **124** may be a portion of the raised wall **80**, the joint boundary wall **124** may both contact the braze joint **106** and partially define the plenum **120**.

[0063] Particularly, inlet channels **118** may be sized and oriented to direct the air **150** in discrete jets to impinge upon the joint boundary wall **124**. The discrete jets of air **150** may impinge (or strike) an impingement surface **126** of the joint boundary wall **124** and cool the impingement surface **126**, which allows for heat transfer from the joint boundary wall **124** and the braze joint **106** to the air. The impingement surface **126** may partially define a boundary of the plenum **120**. For example, the inlet channels **118** may include an exit portion that extends generally perpendicularly to the impingement surface **126** of the joint boundary wall **124**, such that the

discrete jets of air **150** exiting the inlet channels **118** are perpendicular to the surface upon which they strike, e.g., the impingement surface **126** of the joint boundary wall **124**. Once the discrete jets of air have impinged upon the impingement surface **126**, they may be referred to as “post-impingement air” and/or “spent cooling air” because the air has undergone an energy transfer and therefore has different characteristics (e.g., higher temperature and lower pressure than prior to impingement).

[0064] In various embodiments, the plenum **120** may be generally rectangularly shaped in cross-section. For example, the plenum **120** may be bounded by an upstream surface **128** (with respect to the flow of coolant through the cooling circuit **108**), the impingement surface **126** opposite the upstream surface **128**, a radially outer surface **130**, and a radially inner surface **132**. The radially outer surface **130** and the radially inner surface **132** may be spaced apart (e.g., radially spaced apart) and may each extend generally parallel to one another. The impingement surface **126** and the upstream surface **128** may be spaced apart from one another, generally parallel to one another and may each extend generally radially.

[0065] In exemplary embodiments, the cooling circuit **108** may include one or more outlet channels **122** defined in the joint boundary wall **124** (e.g., defined partially in the raised wall **80** and defined partially in the inner wall **110**) and in fluid communication with the plenum **120**. The outlet channel **122** may extend from an inlet **138** defined in the radially outer surface **130** and in fluid communication with the plenum **120** to an outlet **140** defined in the inner surface **136** of the inner wall **110**. The one or more outlet channels **122** may include at least a portion that extends between the impingement surface **126** and the braze joint **106** to cool the braze joint **106** (and the braze material **107**). For example, the one or more outlet channels **122** may include a U-shaped portion **142** and a straight portion **144** (which may extend radially). The U-shaped portion **142** may extend from the inlet **138** defined in the radially outer surface **130** to the straight portion **144**. The straight portion **144** may be disposed axially between the impingement surface **126** and the braze joint **106**, and the straight portion **144** of the outlet channel **122** may extend from the U-shaped portion **142** to an outlet **140** defined in the inner surface **136** of the inner wall **110**. In exemplary implementations, the air **150** exiting the outlet **140** of the outlet channel **122** may film cool the inner surface **136** (and/or the airfoil **56**).

[0066] Referring now specifically to the embodiment shown in FIGS. 7 and 8, the one or more outlet channels **122** and the one or more inlet channels **118** may be alternately arranged. For example, the one or more outlet channels **122** and the one or more inlet channels **118** may not lie in the same plane defined by the radial direction **R** and the direction of the cooling air **150** exiting the inlet channel **118**. Particularly, each outlet channel **122** of the one or more outlet channels **122** may be disposed between two inlet channels **118** of the one or more inlet channels **118**. Similarly, each inlet channel **118** of the one or more inlet channels **118** may be disposed between two outlet channels **122** of the one or more outlet channels **122**. Stated otherwise, the inlet channels **118** and the outlet channels **122** may be spaced apart from one another with respect to a direction **152** that is perpendicular to the cooling air **150** exiting the inlet channel **118**. In some embodiments, the direction **152** may be generally parallel to the upstream surface **128** partially defining plenum **120**. Particularly, the direction **152** may extend along a perimeter of the protrusion **72**. For example, in many embodiments, each outlet channel **122** of the one or more outlet channels **122** may be offset from each inlet channel **118** (e.g., each neighboring inlet channel **118**) of the one or more inlet channels **118** both radially (as shown in FIG. 7) and in the direction **152** that extends along a perimeter of the protrusion **72** (as shown in FIG. 8).

[0067] As shown in FIG. 8, the joint boundary wall **124** may include one or more projections **154** extending from the impingement surface **126**. For example, the projections **154** may extend into the plenum **120** towards the upstream surface **128**. In exemplary embodiments, the outlet channels **122** may be at least partially defined in the projections **154** of the joint boundary wall **124**. Particularly, the straight portion **144** of the outlet channels **122** may be defined in the projections **154** of the

joint boundary wall **124**. Each of the projections **154** may have a semi-circular cross-sectional shape (or other suitable cross-sectional shape). An advantage of the embodiment of FIGS. **7** and **8** is that the impingement surface **126** and the straight portion **144** of the outlet channel **122** may be disposed in closer proximity to the braze joint **106**, thereby improving the cooling of the braze joint **106**.

[0068] As shown in FIG. **9**, in some embodiments, the cooling circuit **108** may include a direct channel **156** extending between an inlet **158** defined in the raised wall **80** (or in the inner wall **110** in other embodiments) to an outlet **160**. In such embodiments, the direct channel **156** may be disposed proximate to the braze joint **106** to provide convective cooling to the braze joint **106**. Additionally, as shown in FIG. **9**, the platform **100** may further include a raised outlet portion **162** extending radially inward from the inner surface **136** of the inner wall **110** of the platform **100** in an area radially inward of the raised wall **80** and proximate to the braze joint **106**. The direct channel **156** may be defined at least partially in the raised outlet portion **162**. For example, the outlet **160** of the direct channel **156** may be defined in the raised outlet portion **162** of the platform **54**. The raised outlet portion **162** may advantageously prevent any melted braze material **107** from wicking or sliding into the direct channel **156** during braze application and brazing. As should be appreciated, the raised outlet portion **162** may be incorporated into one or more of the outlet channels **122** described above with reference to FIGS. **6** through **8**, in order to prevent braze material **107** from entering the cooling circuit **108** during braze application and brazing.

[0069] Referring now to FIGS. **10** through **12**, the cooling circuit **108** may be defined in the protrusion **72** of the airfoil **56** rather than (or in addition to) the cooling circuit **108** being defined in the platform(s) **52**, **54**, in accordance with embodiments of the present disclosure. In such embodiments, the cooling circuit **108** may be disposed in the protrusion **72** proximate the braze joint **106**, to provide cooling (e.g., convective cooling) thereto during operation of the gas turbine **10**. FIG. **10** illustrates a perspective view of an airfoil **56** decoupled from the inner and outer platforms **52**, **54**, in order to show details of the cooling circuit **108**. FIG. **11** illustrates a cross-sectional perspective view of the stator vane **26** having a cooling circuit **108** defined in the protrusion **72**. FIG. **12** illustrates an enlarged cross-sectional view of the stator vane **26** shown in FIG. **11**.

[0070] As shown, the cooling circuit **108** may include a cooling passage **164** defined in the protrusion **72** and extending between an inlet **166** and an outlet **168**. Particularly, the inlet **166** may be defined in a leading edge portion **84** of the protrusion **72**, and the outlet may be defined in a trailing edge portion **86** of the protrusion **72**. In this way, the passage may extend through the pressure side portion **88** (and/or the suction side portion **90**, as shown in FIG. **11**) of the protrusion from the inlet **166** to the outlet **168**. The cooling passage **164** along the pressure side portion **88** may have its own inlet **166** and its own outlet **168** relative to the cooling passage **164** along the suction side portion **90**, or the cooling passages **164** along the pressure side portion **88** and the suction side portion **90** may share a common inlet **166** and/or a common outlet **168**. Additionally, while a single cooling passage **164** is shown along each of the pressure side portion **88** and the suction side portion **90**, it should be understood that more than one cooling passage **164** may be disposed in one or both of the pressure side portion **88** and the suction side portion **90** (e.g., as radially stacked passages).

[0071] While FIGS. **10** through **12** illustrate a cooling passage **164** defined in the protrusion **72** at the tip **70** of the airfoil **56**, it should be appreciated that the cooling passage **164** may alternatively (or additionally) be employed in the protrusion **72** at the base **68** of the airfoil **56**. Because the braze joint **106** bonds the protrusion **72** to the outer platform **54**, and the cooling passage **164** extends through the protrusion **72** proximate the braze joint **106**, the cooling passage **164** may advantageously provide convective cooling to the braze joint **106** during operation of the gas turbine **10**. Any aspects of the cooling passage **164** described above with respect to tip **70** of the airfoil **56** are equally applicable to cooling passage(s) **164** in the base **68** of the airfoil **56**.

[0072] Referring now to FIGS. 13 and 14, cooling circuit 108 may be defined in the protrusion 72 rather than (or in addition to) the cooling circuit 108 being defined in the platform(s) 52, 54, in accordance with embodiments of the present disclosure. In such embodiments, the cooling circuit 108 may be disposed in the protrusion 72 proximate the braze joint 106, to provide convective cooling thereto during operation of the gas turbine 10. FIG. 12 illustrates a perspective view of an airfoil 56 decoupled from the inner platforms 52, 54, in order to show details of the cooling circuit 108, and FIG. 14 illustrates an enlarged cross-sectional view of the stator vane shown in FIG. 13.

[0073] As shown, the cooling circuit 108 may include a cooling passage 174 defined in the protrusion 72 and extending between an inlet 176 and a plurality of outlets 178. Particularly, the inlet 166 may be defined in a leading edge portion 84 of the protrusion 72, and the plurality of outlets 178 may be defined in one of the pressure side wall 64 or the suction side wall 66. In this way, the cooling passage 174 may include a first portion 180 that extends through the pressure side portion 88 (or the suction side portion 90) of the protrusion 72 and a second portion 182 that extends through the pressure side wall 64 (or the suction side wall 66) of the airfoil 56. The second portion 182 of the cooling passage 174 may be oriented generally radially and may be disposed closer to the trailing edge 62 than the leading edge 60 of the airfoil 56.

[0074] As shown in FIG. 14, the plurality of outlets 178 may be radially spaced apart from one another and may extend through the pressure side wall 64 (or the suction side wall 66). In this way, the first portion 180 of the cooling passage 174 may provide convective cooling to the braze joint 106, and the second portion 182 of the cooling passage 174 may provide film cooling to one of the pressure side wall 64 and/or the suction side wall 66 via air 150 exiting the plurality of outlets 178.

[0075] Referring now to FIGS. 15 through 17, the stator vane may further include a cap 184 and an insert 186. The cap 184 may be coupled to the protrusion 72 and may extend across the cavity 92 defined by the protrusion 72 and the airfoil 56. For example, the cap 184 may provide a barrier and/or boundary for air outside of the cavity 92. The cap may be a generally flat plate that is shaped as an airfoil (e.g., an airfoil without a trailing edge). For example, the cap 184 may include a leading edge that contacts the leading edge portion 84 of the protrusion 72, a pressure side that contacts the pressure side portion 88 of the protrusion 72, and a suction side that contacts the suction side portion 90 of the protrusion 72. The insert 186 may be coupled to the protrusion 72 and may extend into the cavity 92. The insert may include a first portion 188 that extends generally parallel to the cap 184 and a second portion 190 that extends generally perpendicularly to the cap 184 (e.g., the second portion 190 may extend radially into the cavity 92).

[0076] In exemplary embodiments, the cap 184 and the insert 186 may define an outlet plenum 192 within the cavity 92 of the stator vane 26. The cooling circuit 108 may be in fluid communication with the outlet plenum 192. For example, as shown in FIGS. 15 through 17, the cooling circuit 108 may be defined in the protrusion 72 rather than (or in addition to) the cooling circuit 108 being defined in the platform(s) 52, 54, in accordance with embodiments of the present disclosure. In such embodiments, the cooling circuit 108 may be disposed in the protrusion 72 proximate the braze joint 106, to provide convective cooling thereto during operation of the gas turbine engine 10. As shown in FIG. 16, the cooling circuit 108 may include an inlet channel 194 and an outlet channel 196. The inlet channel 194 may extend from an inlet 200 defined in a radially outer surface 198 of the protrusion 72, towards the braze joint 106, to the outlet channel 196. The outlet channel 196 may extend from the inlet channel 194 to an outlet 202 defined in an interior surface 204 of the protrusion 72. The outlet 202 may exhaust air from the cooling circuit 108 into the outlet plenum 192.

[0077] In various embodiments, as shown collectively in FIGS. 15 and 16, the protrusion 72 may include a plurality of cooling circuits 108 each in fluid communication with the outlet plenum 192. For example, each cooling circuit 108 may extend from an inlet 200 defined in the radially outer surface 198 of the protrusion 72 to the outlet 202 defined in the interior surface 204 of the protrusion 72. For example, as shown in FIG. 15, the plurality of cooling circuits 108 may be

spaced apart from one another (e.g., equally or unequally spaced apart) around a perimeter of the protrusion **72**. Particularly, the inlets **200** of each cooling circuit **108** of the plurality of cooling circuits **108** may be spaced apart from one another around the perimeter of the protrusion **72**. [0078] In many embodiments, an annular plenum **206** may be defined between the insert **186** and a wall **208** (e.g., the pressure side wall **64** and/or the suction side wall **66**) of the airfoil **56**. For example, the insert **186** may be annular, such that the annular plenum **206** is defined around the entire perimeter of the airfoil **56** between the wall **208** and the insert **186**. In exemplary embodiments, the annular plenum **206** may be fluidly coupled to the outlet plenum **192** via an insert aperture **210**. For example, the insert aperture **210** may be defined in the second portion **190** of the insert **186** to fluidly couple the outlet plenum **192** to the annular plenum **206**, thereby providing air to the annular plenum **206** for cooling the wall **208** during operation of the turbomachine.

[0079] Alternatively, or additionally, as shown in FIG. **17**, the cooling circuit **108** may include a plurality of inlet channels **194**, a plenum **195**, and an outlet channel **196**. A joint boundary wall **211** may be disposed between the plenum **195** and the braze joint **106**, and an impingement surface **212** of the joint boundary wall **211** may partially define the plenum **195**. Each inlet channel **194** may extend from an inlet **202** defined in the interior surface **204** of the protrusion **72** to the plenum **195**, and the outlet channel **196** may extend from the plenum **195** to an outlet defined in the interior surface **204**. Particularly, inlet channels **194** may be sized and oriented to direct the air in discrete jets to impinge upon an impingement surface **212**. The discrete jets of air may impinge (or strike) an impingement surface **212** which allows for heat transfer between the air and the braze joint **106**.

[0080] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims and may include other examples that occur to those skilled in the art. Such other examples are intended to fall within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal language of the claims.

[0081] Further aspects of the invention are provided by the subject matter of the following clauses:

[0082] According to a first aspect, a stator vane comprises: a platform defining an opening; an airfoil having a leading edge, a trailing edge, a suction side wall, and a pressure side wall, the airfoil extending radially between a base and a tip, wherein at least one of the base or the tip includes a protrusion, the protrusion extending into the opening of the platform such that the platform surrounds the protrusion of the airfoil; a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil; and a cooling circuit defined in at least one of the protrusion or the platform to cool the braze joint.

[0083] The stator vane as in any of the preceding clauses, wherein the cooling circuit comprises one or more inlet channels extending to a plenum and configured to direct coolant to impinge upon a joint boundary wall, wherein the joint boundary wall contacts the braze joint and partially defines the plenum.

[0084] The stator vane as in any of the preceding clauses, wherein the cooling circuit further comprises one or more outlet channels defined in the joint boundary wall and in fluid communication with the plenum.

[0085] The stator vane as in any of the preceding clauses, wherein each outlet channel of the one or more outlet channels is offset from each inlet channel of the one or more inlet channels both radially and in a direction that extends along a perimeter of the protrusion.

[0086] The stator vane as in any of the preceding clauses, wherein the platform includes an inner wall, an outer wall, and a raised wall extending between the inner wall and the outer wall, wherein the braze joint is disposed between and fixedly couples the raised wall of the platform to the protrusion of the airfoil.

[0087] The stator vane as in any of the preceding clauses, wherein the one or more inlet channels, the one or more outlet channels, and the plenum are defined in the raised wall and the inner wall of the platform.

[0088] The stator vane as in any of the preceding clauses, wherein the cooling circuit is defined in the protrusion.

[0089] The stator vane as in any of the preceding clauses, wherein the protrusion of the airfoil includes a leading edge portion, a trailing edge portion, a pressure side portion, and a suction side portion, and wherein the cooling circuit comprises a cooling passage extending from an inlet at the leading edge portion to an outlet defined in one of the trailing edge portion, the pressure side wall, or the suction side wall.

[0090] The stator vane as in any of the preceding clauses, wherein the stator vane defines a cavity, wherein an insert is coupled to the protrusion and extends into the cavity, and wherein a cap is coupled to the protrusion and extends across the cavity.

[0091] The stator vane as in any of the preceding clauses, wherein the cap and the insert define an outlet plenum within the cavity of the stator vane, wherein the cooling circuit is in fluid communication with the outlet plenum.

[0092] The stator vane as in any of the preceding clauses, wherein an annular plenum is defined between the insert and a wall of the airfoil, and wherein the annular plenum is fluidly coupled to the outlet plenum via an insert aperture.

[0093] According to another aspect, a turbomachine comprises: a compressor section; a combustion section; and a turbine section comprising at least one stator vane, wherein the at least one stator vane comprises: a platform defining an opening; an airfoil having a leading edge, a trailing edge, a suction side wall, and a pressure side wall, the airfoil extending radially between a base and a tip, wherein at least one of the base or the tip includes a protrusion, the protrusion extending into the opening of the platform such that the platform surrounds the protrusion of the airfoil; a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil; and a cooling circuit defined in at least one of the protrusion or the platform to cool the braze joint.

[0094] The turbomachine as in any of the preceding clauses, wherein the cooling circuit comprises one or more inlet channels extending to a plenum and configured to direct coolant to impinge upon a joint boundary wall, wherein the joint boundary wall contacts the braze joint and partially defines the plenum.

[0095] The turbomachine as in any of the preceding clauses, wherein the cooling circuit further comprises one or more outlet channels defined in the joint boundary wall and in fluid communication with the plenum.

[0096] The turbomachine as in any of the preceding clauses, wherein each outlet channel of the one or more outlet channels is offset from each inlet channel of the one or more inlet channels both radially and in a direction that extends along a perimeter of the protrusion.

[0097] The turbomachine as in any of the preceding clauses, wherein the platform includes an inner wall, an outer wall, and a raised wall extending between the inner wall and the outer wall, wherein the braze joint is disposed between and fixedly couples the raised wall of the platform to the protrusion of the airfoil.

[0098] The turbomachine as in any of the preceding clauses, wherein the one or more inlet channels, the one or more outlet channels, and the plenum are defined in the raised wall and the inner wall of the platform.

[0099] The turbomachine as in any of the preceding clauses, wherein the cooling circuit is defined in the protrusion.

[0100] The turbomachine as in any of the preceding clauses, wherein the protrusion of the airfoil includes a leading edge portion, a trailing edge portion, a pressure side portion, and a suction side portion, and wherein the cooling circuit comprises a cooling passage extending from an inlet at the leading edge portion to an outlet defined in one of the trailing edge portion, the pressure side wall,

or the suction side wall.

[0101] The turbomachine as in any of the preceding clauses, wherein the stator vane defines a cavity, wherein an insert is coupled to the protrusion and extends into the cavity, and wherein a cap is coupled to the protrusion and extends across the cavity.

Claims

1. A stator vane comprising: a platform defining an opening; an airfoil having a leading edge, a trailing edge, a suction side wall, and a pressure side wall, the airfoil extending radially between a base and a tip, the airfoil defining a cavity, wherein at least one of the base or the tip includes a protrusion, the protrusion extending into the opening of the platform such that the platform surrounds the protrusion of the airfoil; a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil; and a cooling circuit defined in the protrusion of the airfoil to cool the braze joint, the cooling circuit fluidly connected to the cavity.
2. The stator vane as in claim 1, wherein the cooling circuit comprises one or more inlet channels and one or more outlet channels.
3. The stator vane as in claim 2, wherein the one or more inlet channels extend from an inlet on a radially outer surface of the airfoil towards the braze joint.
4. The stator vane as in claim 2, wherein the one or more outlet channels extend from the inlet channel away from the braze joint to an outlet on an interior surface of the airfoil.
5. The stator vane as in claim 1, further comprising an insert coupled to the protrusion and extending into the cavity.
6. The stator vane as in claim 5, further comprising a cap coupled to the protrusion and extending across the cavity.
7. The stator vane as in claim 6, wherein the cap and the insert define an outlet plenum within the cavity of the stator vane.
8. The stator vane as in claim 7, wherein the cooling circuit is in fluid communication with the outlet plenum.
9. The stator vane as in claim 8, wherein an annular plenum is defined between the insert and a wall of the airfoil.
10. The stator vane as in claim 9, wherein the annular plenum is fluidly coupled to the outlet plenum via an insert aperture defined in the insert.
11. A turbomachine comprising: a compressor section; a combustion section; and a turbine section comprising a plurality of stator vanes, wherein at least one stator vane of the plurality of stator vanes comprises: a platform defining an opening; an airfoil having a leading edge, a trailing edge, a suction side wall, and a pressure side wall, the airfoil extending radially between a base and a tip, the airfoil defining a cavity, wherein at least one of the base or the tip includes a protrusion, the protrusion extending into the opening of the platform such that the platform surrounds the protrusion of the airfoil; a braze joint disposed between and fixedly coupling the platform and the protrusion of the airfoil; and a cooling circuit defined in the protrusion of the airfoil to cool the braze joint, the cooling circuit fluidly connected to the cavity.
12. The turbomachine as in claim 11, wherein the cooling circuit comprises one or more inlet channels and one or more outlet channels.
13. The turbomachine as in claim 12, wherein the one or more inlet channels extend from an inlet on a radially outer surface of the airfoil towards the braze joint.
14. The turbomachine as in claim 12, wherein the one or more outlet channels extend from the inlet channel away from the braze joint to an outlet on an interior surface of the airfoil.
15. The turbomachine as in claim 11, further comprising an insert coupled to the protrusion and extending into the cavity.
16. The turbomachine as in claim 15, further comprising a cap coupled to the protrusion and

extending across the cavity.

17. The turbomachine as in claim 16, wherein the cap and the insert define an outlet plenum within the cavity of the stator vane.

18. The turbomachine as in claim 17, wherein the cooling circuit is in fluid communication with the outlet plenum.

19. The turbomachine as in claim 18, wherein an annular plenum is defined between the insert and a wall of the airfoil.

20. The turbomachine as in claim 19, wherein the annular plenum is fluidly coupled to the outlet plenum via an insert aperture defined in the insert.
